Research Paper 2: Design and Implementation of Smart Home Energy Management System for Indian Residential Sector

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Abstract

This research presents an IoT-enabled Home Energy Management System designed for the Indian residential sector. The system focuses on monitoring household electricity use and managing loads according to usage patterns. A programmable load-categorization method is introduced to support peak-load management. The study also considers consumer readiness for demand response. The reported results show potential electricity-cost savings of up to 49% under a fixed tariff and 53% under a Time-of-Day tariff.

Introduction

Residential electricity consumption is increasing because homes use more electrical and smart appliances. During peak periods, high demand can increase electricity costs and put additional pressure on the power system. A Home Energy Management System can help by monitoring loads and scheduling flexible appliances. This paper is particularly useful for an Indian project because it considers the residential conditions and tariff environment of India.

Proposed Approach

The system uses IoT technology to monitor and manage household electrical loads. Appliances are categorized according to their characteristics and ability to be shifted or controlled. Loads that do not need to operate at a specific time can be scheduled for more suitable periods. This reduces unnecessary peak demand while maintaining the user's essential requirements.

Load Management

The key idea is programmable load categorization. Essential appliances can continue to operate when required, while flexible appliances can be shifted from high-demand periods to lower-demand periods. Time-of-Day electricity pricing can further encourage users to operate flexible loads when electricity is cheaper. Such scheduling is an important part of smart power management.

Consumer and Practical Considerations

The research also evaluates energy-usage patterns and consumer readiness for demand-response programs. A smart energy system is not useful only because of its technical design; users must be able to understand and accept automatic or scheduled control. Therefore, ease of use, convenience and clear information about energy consumption are important parts of the system.

Results

The study reports cost savings of up to 49% with a fixed tariff and up to 53% with a Time-of-Day tariff. These results indicate that intelligent scheduling and demand-side management can provide meaningful economic benefits. The exact savings depend on appliance usage, tariff structure and user behaviour.

Advantages

- Designed with Indian residential conditions in mind.
- Uses IoT for smart energy management.
- Supports peak-load reduction.
- Allows programmable categorization of loads.
- Can reduce electricity cost through better scheduling.
- Connects technical energy management with user behaviour.

Limitations

The effectiveness depends on correct load information, suitable appliance scheduling and user acceptance. Real-world savings can vary between households because electricity consumption patterns and appliance ownership are different. Internet connectivity and compatible control hardware may also be required.

Conclusion

The paper shows that a smart home energy management system can help Indian households reduce peak demand and electricity cost. Its concept of categorizing appliances and scheduling flexible loads is highly suitable for a student Smart Power Management project. A simplified prototype can use sensors, a microcontroller, relays and a dashboard to demonstrate the same basic idea.

Note: This is an original student-friendly summary of the cited research paper, prepared for SPI academic submission; it is not a reproduction of the original paper.